

Hospital Operations & Patient Analysis

Final Project Documentation

Infosys Springboard Virtual Internship 7.0

Academic Year: 2026

Internship Domain: Data Analysis and Visualization

Project Area: Healthcare Operations Analytics

Tools: Tableau, Power BI, Python, Pandas, Jupyter

Guide/Mentor: Pathina Likhita

Prepared By: Rutuja Warbhog

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1. PROJECT OVERVIEW

The Hospital Operations and Patient Analysis project is a healthcare analytics and visualization project developed to convert hospital operational and patient-level data into clear and useful management insights. The project analyzes admissions, discharges, patient transfers, department performance, billing and revenue, readmissions, bed and ICU capacity, staffing, and equipment utilization.

The main purpose of the project is to provide a single analytical view of hospital operations using interactive dashboards. Instead of reviewing large numbers of records manually, users can compare hospitals, departments, locations and time periods through consistent KPIs, filters and visualizations. The project is intended for descriptive operational analysis and decision support, not for clinical diagnosis or treatment decisions.

2. PROBLEM STATEMENT AND OBJECTIVES

Problem Statement

Hospital datasets contain many operational records and measures. When admissions, patient movement, revenue, capacity, staffing and resource information are reviewed separately, it becomes difficult to understand the complete operational situation. Manual review also makes it harder to compare departments, identify trends, monitor utilization and verify important KPIs. This project addresses the problem by cleaning and preparing the data, defining validated KPIs, and presenting the results through four connected dashboards.

Project Objectives

- Analyze hospital admission, discharge and patient-flow trends.
- Compare department-wise workload, capacity, staffing and operational performance.
- Analyze revenue, readmission, length of stay and patient-segment patterns.
- Monitor beds, ICU capacity, staff utilization and equipment utilization.
- Develop four interactive dashboards with filters, navigation and chart interactions.
- Validate major KPI calculations independently using Python and Pandas.
- Test dashboard filters, navigation, interactions and data consistency before final delivery.

3. TOOLS AND TECHNOLOGIES

Tool / Technology	Usage in the Project
Python	Data preparation, cleaning, transformation and validation.
Pandas	Data manipulation, grouping, calculations and independent KPI validation.
NumPy	Numerical calculations used during data preparation and analysis.
Jupyter Notebook	Used for the data-cleaning and KPI-validation workflow.
Tableau Desktop	Main platform used to develop the interactive dashboards and final workbook.
Power BI	Additional visualization tool included in the project scope.
CSV / Excel	Used to store raw, cleaned and analytical datasets.
GitHub	Version control, project storage, documentation and final delivery.

4. PROJECT METHODOLOGY / MODULES

The project follows an end-to-end data analytics workflow from raw data to validated dashboards and final documentation.

- 1. Data Collection:** Collect and store the raw hospital dataset in CSV format.
- 2. Data Cleaning and Preparation:** Check field names, data types, missing values, duplicates, categorical values and date consistency.
- 3. Derived Field Creation:** Create analytical fields such as Age Group, Admission Month, Admission Year, Length of Stay category, department capacity and staff-utilization measures.
- 4. KPI Generation:** Calculate important hospital, patient-flow, department and resource KPIs using defined formulas and aggregation rules.
- 5. Dashboard Development:** Build the Hospital Overview, Patient Flow, Department Analysis and Resource Utilization dashboards in Tableau.
- 6. Testing and Validation:** Compare important Tableau results with independent Python/Pandas calculations and test filters, navigation and interactions.
- 7. Documentation and Delivery:** Prepare the final documentation, methodology, KPI definitions, dashboard guide and project repository.

5. DATASET DESCRIPTION

The cleaned working dataset contains 10,000 records and 53 fields. The fields cover hospital details, patient information, admissions, clinical/service information, financial values, staffing and capacity, equipment information and transfer activity.

Data Area	Important Fields
Hospital	Hospital ID, Hospital Name, Hospital Type, City, State
Patient	Patient ID, Patient Name, Gender, Age, Blood Type
Admissions	Admission Date, Discharge Date, Admission Type
Clinical / Service	Diagnosis, Treatment, Medication, Test Result
Financial	Billing Amount, Insurance Provider
Resources	Beds, ICU Beds, Doctors, Nurses, Staff Count
Equipment	Equipment ID, Equipment Status, Usage Duration
Transfers	Transferred, Transfer From/To Department, Transfer Date

Data Cleaning Performed

- Checked field names, dataset structure, data types and overall completeness.
- Reviewed missing values, duplicate records and key identifiers.
- Standardized categorical values such as departments and operational indicators.
- Validated admission and discharge dates and retained same-day stays as 0 days.
- Recalculated Length of Stay as Discharge Date minus Admission Date.
- Created analytical fields such as Age Group, Admission Month/Year, LOS Category, department bed capacity, ICU capacity and staff utilization.
- Kept patient, hospital, department and equipment data grains separate where necessary to avoid incorrect aggregation.

Validated Length of Stay Statistics

Statistic	Validated Value
Mean Length of Stay	78.78 days
Median Length of Stay	44 days
Maximum Length of Stay	338 days
Records with LOS > 30 days	5,497
Records with LOS > 90 days	3,706

6. IMPORTANT KPIS

The following KPI values were independently validated using Python/Pandas and compared with the Tableau dashboard. Display rounding is used in some dashboard cards.

KPI	Validated Value	Definition / Basis
Total Admissions	10,000	Total admissions in the analyzed dataset.
Total Revenue	Rs. 2.50 Billion	Sum of Billing Amount.
Average Length of Stay	78.78 days	Average recalculated Length of Stay.
Average Age	45.30 years	Average patient age.
Readmission Rate	49.87%	Readmitted patients divided by total patients.
Bed Utilization / Occupancy Rate	36.76%	Average of the prepared Bed_Occupancy_Rate_% field.
Staff Utilization Rate	57.92%	Average of the prepared Staff_Utilization_%_Derived field.
Patients Transferred	1,760	Patients marked Transferred = Yes.
Transfer Rate	17.60%	Transferred patients divided by total patients.
Average Daily Admissions	19.05	Admissions divided by distinct admission dates.
Equipment Utilization Rate	33.55%	Equipment marked In Use divided by observed equipment.
Average Department Bed Capacity	491.69	Average department-level bed capacity.
Average ICU Capacity	98.70	Average department ICU bed capacity.
Efficiency Score	47.90%	Weighted operational comparison score.

Efficiency Score Formula

Efficiency Score = (Bed Utilization Rate x 0.40) + (Staff Utilization Rate x 0.40) + ((1 - Readmission Rate) x 0.20). The validated overall score is approximately 47.90%. This is an operational comparison measure and should not be treated as a clinical quality score.

Important note: the 36.76% bed KPI is based on the average of the prepared Bed_Occupancy_Rate_% field. A separate count-based calculation using Bed Occupied = Yes gives a different result, so the metric definition and denominator must remain consistent when reporting the KPI.

7. DASHBOARD EXPLANATION AND SCREENSHOTS

The project contains four integrated dashboards. Each dashboard answers a different operational question while using the same cleaned dataset and consistent filters.

7.1 HOSPITAL OVERVIEW DASHBOARD

Purpose

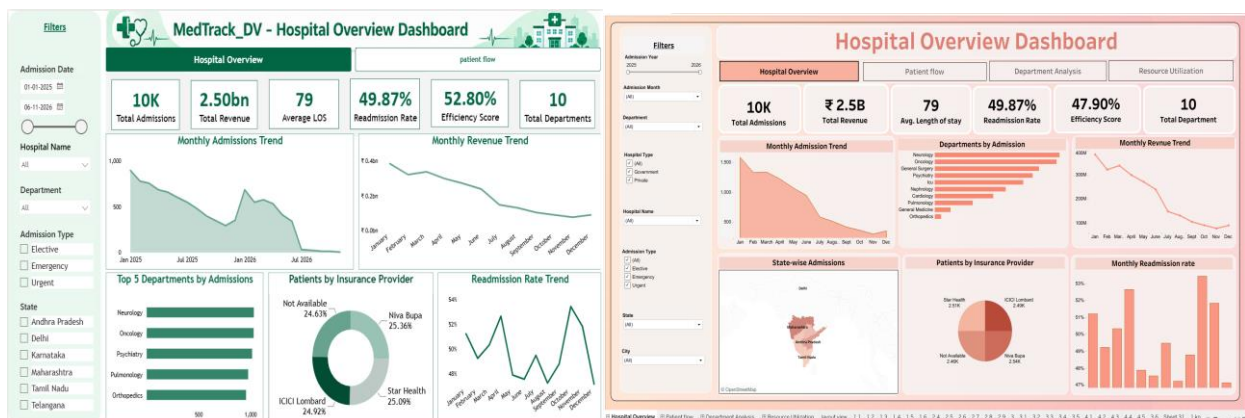
The Hospital Overview dashboard provides a high-level view of hospital activity and performance. It is the starting point for reviewing admissions, revenue, length of stay, readmissions, efficiency and overall trends before moving to detailed dashboards.

Important KPIs / Views

- Total Admissions: 10,000
- Total Revenue: Rs. 2.50 Billion
- Average Length of Stay: 78.78 days
- Readmission Rate: 49.87%
- Efficiency Score: 47.90%
- Monthly admission and revenue trends, state-wise admissions and insurance-provider analysis

Observations

- The dataset represents 10,000 admissions across the selected analytical scope.
- The readmission rate is close to half of the analyzed patient population, so it is an important operational metric for further investigation.
- Average Length of Stay is 78.78 days, indicating that long and extended stays have a strong effect on overall operations.



7.2 PATIENT FLOW DASHBOARD

Purpose

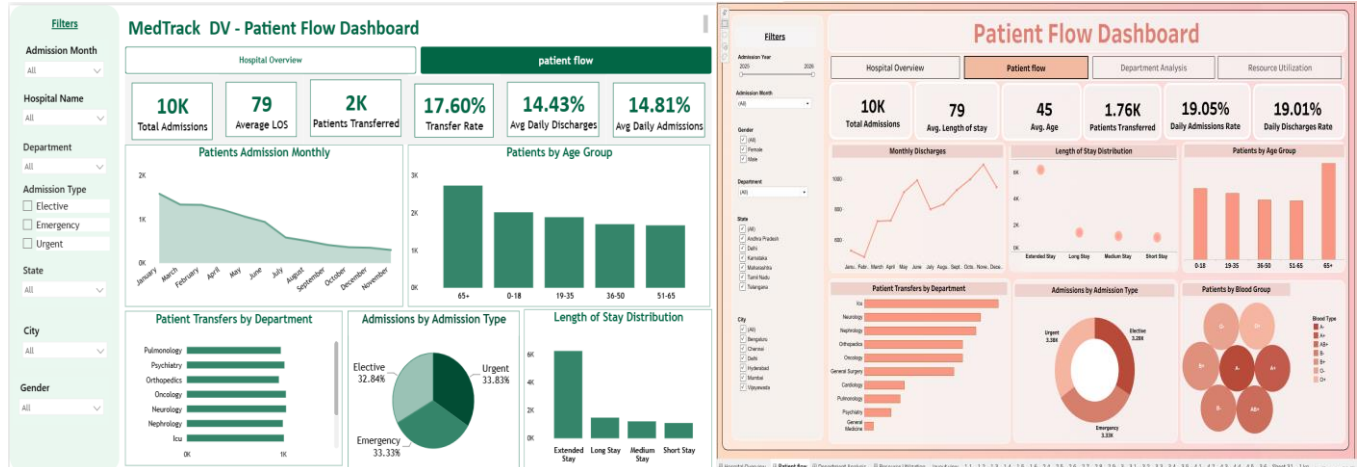
The Patient Flow dashboard explains who the patients are and how they move through the hospital. It focuses on admissions, discharges, transfers, admission types, age groups and Length of Stay categories.

Important KPIs / Views

- Total Admissions: 10,000
- Patients Transferred: 1,760
- Transfer Rate: 17.60%
- Average Daily Admissions: 19.05
- Admission Types: Urgent 3,383; Emergency 3,333; Elective 3,284
- LOS Categories: Short 1,092; Medium 1,208; Long 1,474; Extended 6,226

Observations

- Approximately 17.60% of patients are marked as transferred.
- Admission types are distributed relatively evenly across Urgent, Emergency and Elective categories.
- Extended Stay is the largest LOS category, with 6,226 records, making it a major operational pattern in the dataset.



7.3 DEPARTMENT ANALYSIS DASHBOARD

Purpose

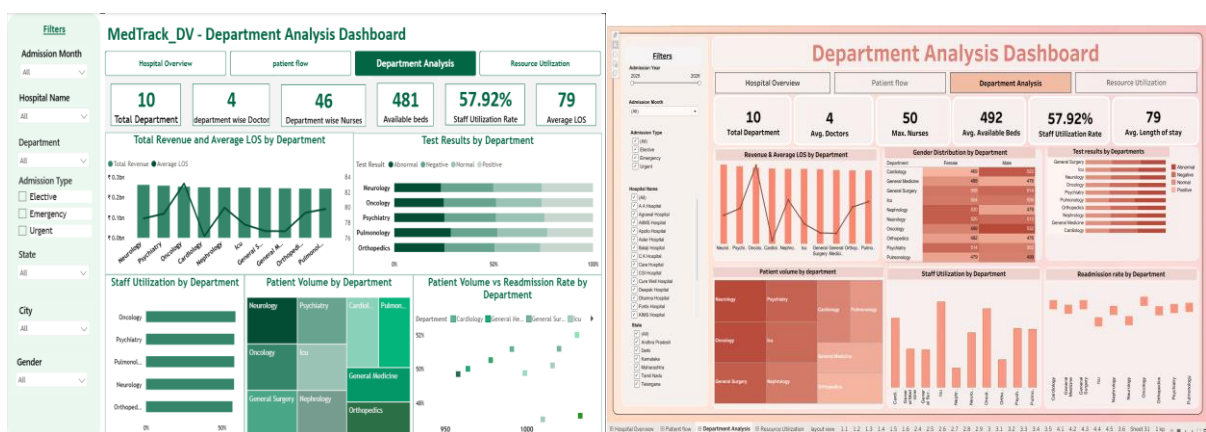
The Department Analysis dashboard compares workload and performance across departments using patient volume, revenue, Length of Stay, staff utilization, bed occupancy, test results, readmissions and department capacity.

Important KPIs / Views

- Total Departments: 10
- Average Doctors: approximately 4
- Maximum Nurses: 50
- Average Department Bed Capacity: 491.69
- Admissions, revenue, LOS, staff utilization, bed occupancy and readmission by department

Observations

- Neurology has the highest admission volume, while Orthopedics has the lowest.
- Oncology has the highest average Length of Stay.
- ICU has the highest staff utilization, while Psychiatry has the highest average bed occupancy among departments.



7.4 RESOURCE UTILIZATION DASHBOARD

Purpose

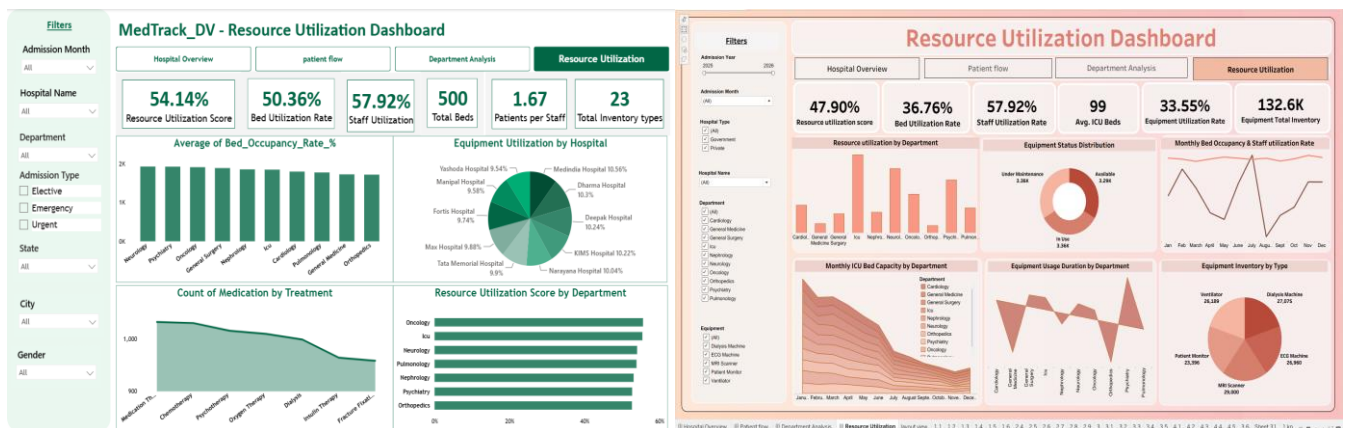
The Resource Utilization dashboard focuses on whether beds, staff, ICU capacity and equipment are available and utilized effectively. It supports operational monitoring by combining utilization measures with capacity and equipment-status information.

Important KPIs / Views

- Bed Utilization / Occupancy Rate: 36.76%
- Staff Utilization Rate: 57.92%
- Average ICU Capacity: 98.70
- Equipment Utilization Rate: 33.55%
- Efficiency / Resource Score: 47.90%

Observations

- Staff utilization is higher than bed and equipment utilization in the overall dataset.
- Equipment utilization is 33.55%, so equipment status should be reviewed together with demand and availability.
- Resource measures should be interpreted together with admissions, Length of Stay, capacity and department context rather than as isolated values.



8. TESTING AND VALIDATION

Testing was performed to verify KPI accuracy, data consistency, dashboard usability and interaction behavior. Independent calculations were created using the cleaned hospital dataset and Pandas, and the results were compared with Tableau values. Minor display-rounding differences were accepted where the underlying values matched.

Testing Areas

- KPI calculations and displayed dashboard values
- Filters for hospital, state, city, department, admission type and time period
- Navigation between the four dashboards
- Chart interactions and cross-filtering
- Tooltips and dashboard layout
- Admission and discharge date consistency
- Duplicate checks and chart/data consistency

Testing Outcome

Testing Metric	Result
Total Test Cases	40
Passed	39
Requires Fix / Review	1
Pass Rate	97.5%
Major Issues	0
Overall Result	PASS WITH MINOR FIXES

The remaining review item is the Total Inventory KPI in the Resource Utilization dashboard. The metric definition and displayed value should be confirmed and revalidated before using it as an official final KPI. Other previously validated major KPIs match the independent calculations.

9. RESULTS, KEY INSIGHTS, LIMITATIONS AND CONCLUSION

Results and Key Insights

- A complete four-dashboard analytical solution was developed using a cleaned dataset of 10,000 records and 53 fields.
- The project provides validated operational KPIs for admissions, revenue, LOS, readmissions, transfers, bed utilization, staff utilization, equipment utilization and efficiency.
- Neurology records the highest admission volume, while Orthopedics records the lowest.
- Extended Stay is the largest Length of Stay category.
- Staff utilization is 57.92%, bed utilization/occupancy is 36.76% and equipment utilization is 33.55%.
- Approximately 17.60% of patients are marked as transferred.

Limitations

- The dataset is an analytical extract and is not connected to a live hospital information system.
- Several capacity and utilization fields are derived measures and should be verified against official source-system definitions before production use.
- Different data grains such as patient, department, hospital and equipment data should not be combined without checking for duplication.
- The dashboards provide descriptive operational analysis and do not establish clinical causes or replace clinical judgment.
- The Resource Utilization Total Inventory KPI still requires final definition/value revalidation.

Future Improvements

- Automated data refresh and scheduled dashboard updates.
- Hospital-level benchmarking and department performance comparison.
- Forecasting of admissions, bed demand and resource requirements.
- Capacity-planning and staffing-support analysis.
- Additional automated validation checks before dashboard publication.

Conclusion

The Hospital Operations and Patient Analysis project demonstrates a complete analytics workflow from data collection and cleaning to KPI engineering, dashboard development, testing and final documentation. The four dashboards - Hospital Overview, Patient Flow, Department Analysis and Resource Utilization - provide complementary views of hospital performance and support evidence-based operational analysis. The project also demonstrates the importance of independent KPI validation, clear metric definitions and consistent filtering when presenting healthcare operational data.

10. GITHUB / PROJECT LINK AND FOLDER STRUCTURE

GitHub Repository-

<https://github.com/springboardmentorpl/Hospital-Operations-and-Patient-Analytics-Dashboard-Group-2/tree/Rutuja-Manishkumar-Warbhog>

The repository contains the raw and cleaned datasets, data-preparation and validation scripts, project documentation and the Tableau dashboard workbook.

Project Location / File	Purpose
data/Hospital_RawDataset_Updated.csv	Raw input dataset.
data/hospital_cleaned.csv	Cleaned and enriched dataset used for analysis and Tableau.
scripts/	Data preparation, cleaning and KPI validation scripts.
docs/	Project documentation, KPI definitions, methodology and validation material.
dashboard/Hospital Operations and Patient Analysis.twb	Tableau workbook containing the dashboard solution.
Final Project Package	Four dashboards, KPI definitions, dashboard guide, methodology and QA/testing documentation.

Final Deliverables

- Cleaned hospital dataset
- Data-preparation and validation scripts
- Tableau workbook
- Four integrated dashboards
- KPI definitions
- Dashboard guide and healthcare methodology
- Testing and validation documentation
- Final internship project documentation